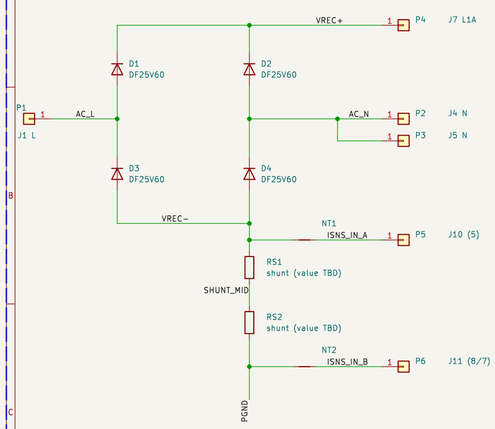
LeafOBC-WafflePlate: first-pass schematic of the 2012 Nissan Leaf OBC power module ("waffle plate")

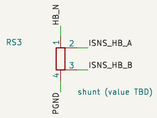
DRAFT. Drawing style does not indicate verification: see each part's Verified field (sections A, D and the PFC driver are meter-verified). Topology is from the community draft schematic plus identified part numbers. Connections are NOT yet verified against the module. Hidden fields per part: Verified, Draft_ref, Location_px (pixel position in images/stitched/waffle-plate-top.jpg).

A. Input bridge and shunts (upper left)



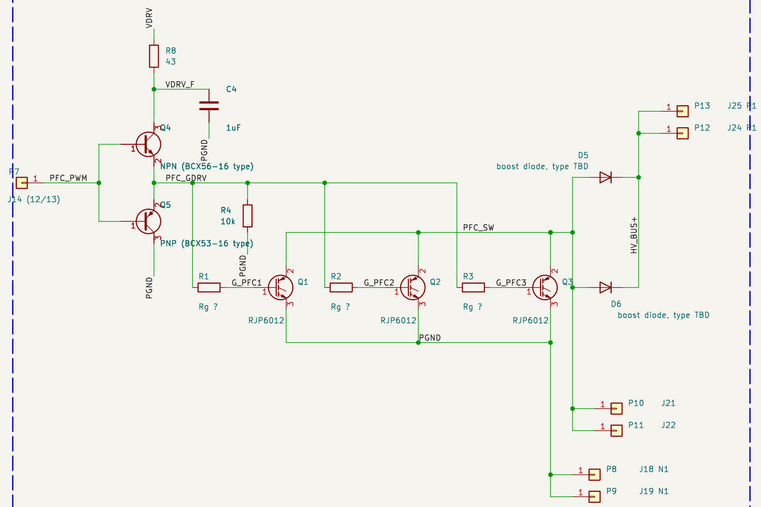
RS1 + RS2: one current shunt in two parts.

Kelvin sense taken from the outer terminals (NT1, NT2).



RS3: full-bridge return (section C)

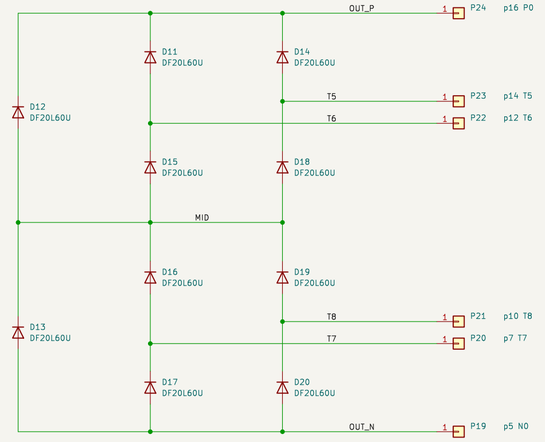
B. PFC boost stage (mid left)



Q1-Q3 in parallel, each with its own gate resistor. D5, D6 in parallel.

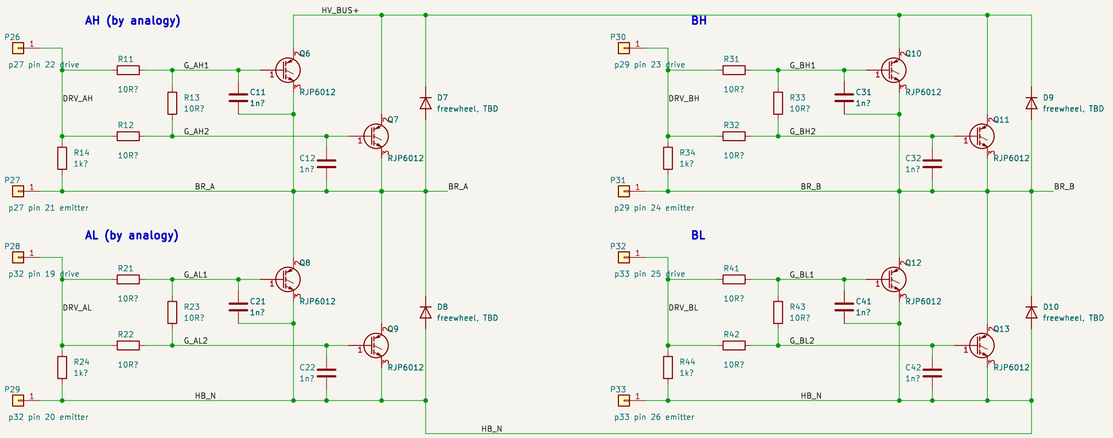
Gate driver (Q4, Q5, R4, R8, C4) meter-verified. IGBTs, gate resistors and boost diodes believed correct, not yet meter-verified.

D. Output rectifier: two series-stacked bridges (upper right)



Verified with meter: D11-D20 and pads.

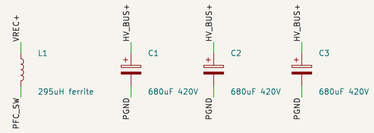
C. Primary full bridge: legs A and B, each switch 2 IGBTs in parallel + gate network (lower right)

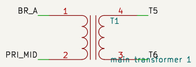


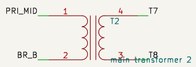
Gate networks: BH/BL traced from copper, AH/AL by analogy. Values with ? are reference values (Ukrainian schematic and replica), unconfirmed.

Drive pads AL/BL from the Ukrainian schematic. AL (Q8, Q9) failed on the reference module. None of section C is meter-verified. See docs/full-bridge-gate-drive.md

E. Off-module parts (charger PCB)

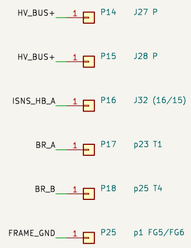






Primaries shown in series: verify.

F. Remaining module pads (draft numbering, unverified)



Pads for A, B and D are drawn next to their circuit.

Unassigned draft pads: J8 (6, IC312_PFC), J12/J13 (11), J15 (10/9), J30/J31 (N-N1), p31 (17/18, thermistor TH202, not yet drawn)

CERN-OHL-S-2.0
Sections A and D meter-verified. Rest unverified.
LeafOBC-WafflePlate

Sheet: /
File: waffle-plate.kicad_sch

Title: Leaf OBC waffle plate: original module

Size: A2	Date:	Rev: 0.1-draft
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1